Practice Quiz: Cognition

Part A: Multiple Choice

1. What is the primary role of **Cognition** in Active Inference? A) To maximize reward B) To minimize variational free energy C) To increase entropy D) To eliminate the Markov Blanket

2. In Robotics, Cognition is best described as: A) A static property B) A dynamic process C) An external state D) A random variable

3. Which mathematical quantity is most central to Cognition? A) The Lagrangian B) The expected free energy C) The surprisal D) The precision

4. How does Cognition relate to the concept of the Generative Model? A) It is separate from the model B) It is a component of the model C) It destroys the model D) It is only relevant for the environment

5. A failure in Cognition would likely result in: A) Perfect prediction B) Generalized surprise C) Immediate death of the agent D) Zero entropy

6. Which scale is most relevant for analyzing Cognition in this course? A) Quantum B) Neural C) Social D) All of the above

7. Cognition connects directly to which other component? A) The step before it B) The step after it C) Both A and B D) None of the above

Part B: Short Answer

1. Explain how **Cognition** facilitates the minimization of prediction error.
2. Provide a concrete example of Cognition failing in a Robotics scenario.
3. How would you model Cognition using a POMDP (Partially Observable Markov Decision Process)?